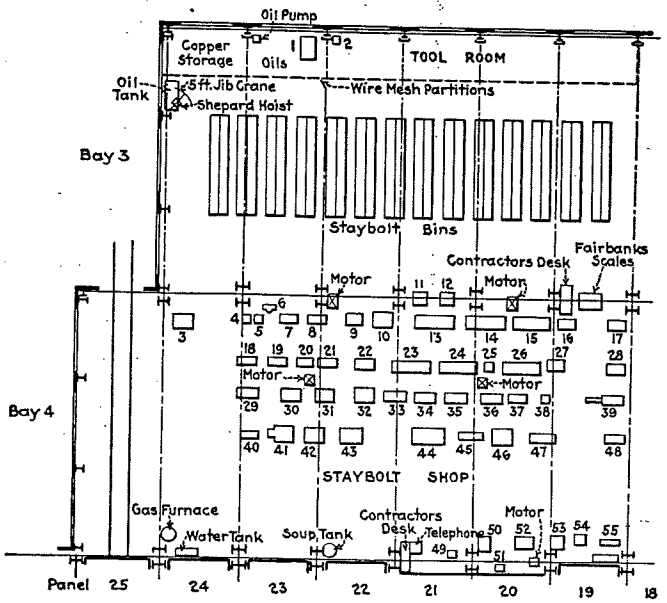


Machinery Located in the Staybolt Shop and Tool Room

1—Brown & Sharp No. 2—30-inch universal grinder, motor on steel work overhead, belt drive.
2—Willey 12-inch, double emery wheel, motor on machine, direct drive.
3—Hilles & Jones 1¼-inch bar shear, motor on machine, belt drive.
4—Harrington centering machine, belt drive from line shaft.
5—Baldwin ⅝ to 1¾-inch centering machine, belt drive from line shaft.
6—Baldwin double emery wheel, belt drive from line shaft.
7—Harrington 4-spindle drill press, belt drive from line shaft.
8—Harrington 4-spindle multiple drill, belt drive from line shaft.
9—Ferracute Machine Company special squaring press, belt drive from line shaft.
10—Ferracute Machine Company special squaring press, belt drive from line shaft.
11—Landis double-spindle, staybolt threading machine, motor on machine, direct drive.
12—Landis double spindle, staybolt threading machine, motor on machine, direct drive.
13—Harrington 6-spindle staybolt threader, belt drive from line shaft.
14—Harrington 6-spindle staybolt threader, belt drive from line shaft.
15—Baldwin 6-spindle staybolt threader, belt drive from line shaft.
16—Bement-Miles double staybolt cutter, belt drive from line shaft.
17—Bement-Miles double staybolt cutter, belt drive from line shaft.
18—Harrington 4-spindle multiple drill, belt drive from line shaft.
19—Harrington 4-spindle multiple drill, belt drive from line shaft.
20—Harrington 4-spindle multiple drill, belt drive from line shaft.
21—Harrington 2-spindle multiple drill, belt drive from line shaft.
22—Pratt & Whitney ⅝ to 1¼-inch double rounder, belt drive from line shaft.
23—Harrington 6-spindle staybolt threader, belt drive from line shaft.
24—Harrington 6-spindle staybolt threader, belt drive from line shaft.
25—Baldwin 10-inch emery wheel, belt drive from line shaft.
26—Harrington 6-spindle staybolt threader, belt drive from line shaft.
27—Bement-Miles staybolt cutter, belt drive from line shaft.
28—Bement-Miles staybolt cutter, belt drive from line shaft.
29—Harrington 16-inch by 3-foot rounder, belt drive from line shaft.
30—Acme staybolt turner, belt drive from line shaft.
31—Acme 1½-inch double staybolt cutter, belt drive from line shaft.
32—Acme 1½-inch double staybolt cutter, belt drive from line shaft.
33—Harrington 16-inch crown-bolt lathe, belt drive from line shaft.
34—Harrington 16-inch by 30-inch crown-bolt lathe, belt drive from line shaft.
35—Harrington 19-inch crown-bolt lathe, belt drive from line shaft.
36—Harrington 16-inch by 36-inch crown-bolt lathe, belt drive from line shaft.
37—Acme single staybolt cutter, belt drive from line shaft.
38—Baldwin 8-inch emery wheel, belt drive from line shaft.
39—Warner-Swasey No. 6 turret lathe, motor on machine, direct drive.
40—Acme 1-inch rounder, belt drive from line shaft.
41—Acme 1½-inch staybolt turner, belt drive from line shaft.
42—Baldwin double staybolt cutter, belt drive from line shaft.
43—Acme double staybolt cutter, belt drive from line shaft.
44—Hartness (Jones & Lamson) 3-inch by 36-inch turret lathe, motor on machine, direct drive.
45—Lodge & Davis 16-inch lathe, belt drive from line shaft.
46—Acme 1½-inch double staybolt cutter, belt drive from line shaft.
47—Cregar 2-inch single staybolt cutter, belt drive from line shaft.
48—Cooper, Jones & Company forming lathe, motor on machine, direct drive.
49—Baldwin single staybolt saw, motor on building column, belt drive.
50—Hilles & Jones No. 0 staybolt shear, belt drive from line shaft.
51—J. G. Blount, 6-inch twist drill grinder, motor drive.
52—Baldwin double, circular staybolt saw, belt drive from line shaft.
53—Hendey shaper, belt drive from line shaft.
54—Landis chaser grinder, motor on machine, direct drive.
55—Bradford 16 by 72-inch lathe for radial staybolts, belt drive from line shaft.

smith shop located elsewhere in the plant. The bar is formed by heating and shaping the ends in upsetting machines. The head end is formed with a rough button or curved head, straight or tapered throat and a square lug. The other end has a straight upset for the length of the thread.

The first operation after receipt from the blacksmith shop is the rounding of the point. This is done in one of two rounders located in panel 23 and designated as Nos. 29 and 40 on the shop layout plan on this page. These machines are similar to that used for waterspace staybolts except that the bolt is gripped in the mid-length by double-V clamps.

High skids are located between these machines and staybolt-turning machines located adjacent to the rounders in the same panel. These skids serve as a table on which the bolts are passed between one machine and another.

The staybolt cutters involved in the second operation are used for roughing. They are double-spindle machines fitted with a special-shaped cutter ground to the proper throat taper and underside of head. Stops are set on the machine to ensure the proper diameter of staybolt at the throat and to limit the travel of the cutter in way of the head. By this means production is increased, each machine being operated by one or two operators, each operating one spindle.

The bolts are transferred by skid to one of two special lathes located in panel 21. These lathes, developed by The Baldwin Locomotive Works, have a special die head attached to the carriage so that when the bolt is once in the lathe, it is not removed until all of the oper-